



Lecture 15

Sampling

Announcements

- **HW 5** due today @ 5pm
 - **Project 1** due Friday 2/23 @ 5pm
 - **Lab 5** due Friday 2/23 @ 5pm
 - **Midterm** on March 8th, 7pm - 9pm
 - Scope: everything up to and including lecture that day
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Sampling

Random Samples

- **Population:**
 - Set of all elements from whom a sample will be drawn
- **Deterministic sample:**
 - Sampling scheme doesn't involve chance
- **Random sample:**
 - A sample for which it is possible to calculate, before the sample is drawn, the chance with which any subset of elements will enter the sample
 - Not all individuals / groups have to have equal chance of being selected

(Demo)

Sample of Convenience

- Example: sample consists of whoever walks by
 - Just because you think you're sampling "randomly", doesn't mean you have a random sample.
 - If you can't figure out ahead of time
 - what's the population
 - what's the chance of selection, for each group in the population
- then you don't have a random sample
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Distributions

Probability Distribution

- Random quantity with various possible values
 - “Probability distribution”:
 - All the possible values of the quantity
 - The probability of each of those values
 - If you can do the math, you can work out the probability distribution without ever simulating it
 - But... simulation is often easier!
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Empirical Distribution

- “Empirical”: based on observations
- Observations can be from repetitions of an experiment
- “Empirical Distribution”
 - All **observed** unique values
 - The proportion of times each value appears

(Demo)

Large Random Samples

Law of Averages / Law of Large Numbers

If a chance experiment is repeated **many times**, **independently** and **under the same conditions**, then the **proportion of times that an event occurs** **gets closer** to the **theoretical probability of the event**

As you increase the number of rolls of a die, the proportion of times you see the face with five spots gets closer to $1/6$

Empirical Distribution of a Sample

- If the sample size is **large**,
- then the **empirical distribution of a uniform random sample resembles the distribution of the population**,
- with **high probability**

(Demo)

A Statistic

Inference

- **Statistical Inference:**

Making conclusions (about the population) based on data in random samples

fixed

- **Example:**

Use the data to guess the value of an unknown number

depends on the random sample

Create an **estimate** of the unknown quantity

Terminology

- **Parameter**
 - A number associated with the population
- **Statistic**
 - A number calculated from the sample

A statistic can be used as an **estimate** of a parameter

(Demo)

Probability Distribution of a Statistic

- Values of a statistic vary because random samples vary
 - “Sampling distribution” or “probability distribution” of the statistic:
 - All possible values of the statistic,
 - and all the corresponding probabilities
 - Can be hard to calculate
 - Either have to do the math
 - Or have to generate all possible samples and calculate the statistic based on each sample
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Empirical Distribution of a Statistic

- Empirical distribution of the statistic:
 - Based on simulated values of the statistic
 - Consists of all the observed values of the statistic,
 - and the proportion of times each value appeared
 - Good approximation to the probability distribution of the statistic
 - if the number of repetitions in the simulation is large
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